class notes

2026-05-16

Photosynthesis: How Plants Make Their Own Food

Learning objectives

- Identify the location of photosynthesis in plants.
- Explain the role of chlorophyll in capturing sunlight.
- Describe the three ingredients needed for photosynthesis.
- Explain the two stages of photosynthesis: light reaction and Calvin cycle.
- Predict the importance of photosynthesis for the Earth's atmosphere.

Key concepts

Chloroplasts

- Chloroplasts are small, green structures within plant cells.
- They contain chlorophyll, which is essential for photosynthesis.
- Chloroplasts are primarily found in the leaves of plants.

Chlorophyll

- Chlorophyll is a green pigment that captures energy from sunlight.
- This captured energy is used to drive the process of photosynthesis.
- Chlorophyll is responsible for the green color of most plants.

The Light Reaction and Calvin Cycle

- The light reaction uses sunlight to split water molecules and produce energy carriers.
- The Calvin cycle uses the energy from the light reaction to combine carbon dioxide with hydrogen, building glucose.
- The Calvin cycle does not directly require light, but depends on the energy produced in the light reaction.

Worked example

- A maple tree absorbs water through its roots and takes in carbon dioxide through its stomata. The light reaction in the chloroplasts within the leaves uses sunlight to produce glucose, a sugar that the tree uses for energy. This process relies on chlorophyll to capture the sunlight energy.

Common misconceptions

- Plants do not "eat" the soil. They take in water and minerals from soil, but their food (glucose) is made inside the leaves.
- Trees do not produce their oxygen at night. The light reaction only runs in sunlight; at night, plants release a small amount of carbon dioxide as they respire.
- Chlorophyll is not the only pigment in plants. Leaves contain other pigments, which is why some turn red, orange, or yellow in autumn.